BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: MRD-IX-21(A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-IX-21(A)

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Which bio element makes most of the composition of organisms body?

(A) Nitrogen
(B) Oxygen
(C) Carbon
(D) Hydrogen

Q2. Deductions are made from:

(A) Observations
(B) Hypothesis
(C) Experiment results
(D) Solution of biological problem

Q3. Maximum biodiversity is found in:

(A) Mountains
(B) Deserts
(C) Grasslands
(D) Forests

Q4. The organelle which provides energy to the cell is:

(A) Golgi apparatus
(B) Ribosome
(C) Mitochondria
(D) Nucleus

Q5. In plants which tissue makes new cells?

(A) Meristematic Tissue
(B) Collenchyma tissue
(C) Epidermal tissue
(D) Ground tissue

Q6. The type of cell division which maintains the chromosome number in daughter cells is:

(A) Mitosis
(B) Meiosis
(C) Both mitosis and meiosis
(D) None of these

Q7. Meiosis is also known as:

(A) Mitotic division
(B) Somatic division
(C) Reduction division
(D) All these

Q8. One of the given may function as coenzyme for the enzymes.

(A) Proteins
(B) Carbohydrates
(C) Vitamins
(D) DNA

Q9. Photo synthesis takes place in:

(A) Mesophyll cells
(B) Guard cells
(C) Phloem cells
(D) All these

Q10. Which of the given may not be required for respiration?

(A) Oxygen
(B) Glucose
(C) Enzymes
(D) CO_2

Q11. Which substance is used by plants to make proteins?

(A) Carbondioxide
(B) Oxygen
(C) Nitrates
(D) Vitamins

Q12. The exchange of materials between the blood and tissues occurs in:

(A) Capillaries
(B) Veins
(C) Arteries
(D) Arteries and veins

Section - B

Time Allowed: 2:40 Hours | **Maximum Marks:** 32

QII. Write short answers of any eight (8) of the following parts. All parts carry equal marks.

1. What are the main divisions of biology?
2. Write the role of hypothesis in biological method.
3. What are the aims of classification?
4. Differentiate between active and passive transport with examples.
5. Draw label diagram of mitochondria.
6. Define cell cycle. Name events of interphase.

7. Differentiate between anabolism and catabolism.

8. Write short note on ATP.

9. What is lactic acid fermentation?

10. Write any four advantages of organic fertilizers.

11. What is the significance of transpiration?

Section - C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks.

QIII.

(a) [4] Discuss multi-cellular organization

(b) [3] What is binomial nomenclature?

QIV.

(a) [4] Define tissue. Describe types of compound tissues in plants.

(b) [3] Write significance of mitosis.

QV.

(a) [4] What are enzymes? Discuss Lock and Key Model of enzymes action.

(b) [3] Photosynthesis is a life sustaining process. Elaborate the statement.

QVI.

(a) [4] Draw label diagram of human digestive system.

(b) [3] Differentiate between arteries and veins.